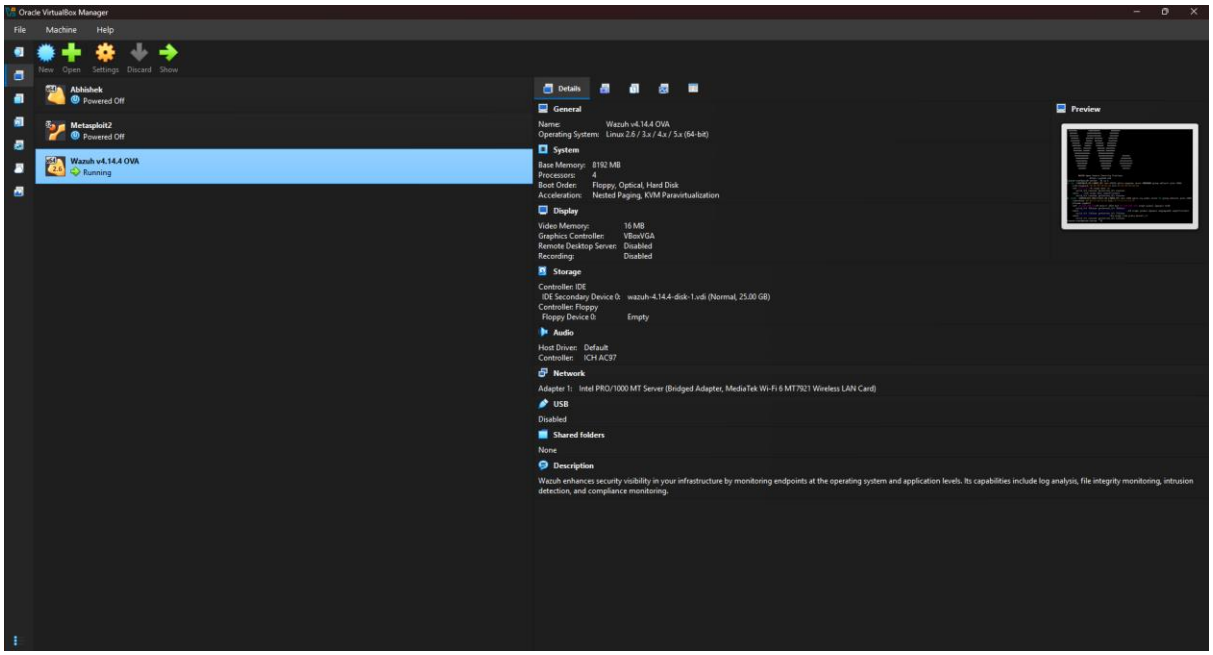
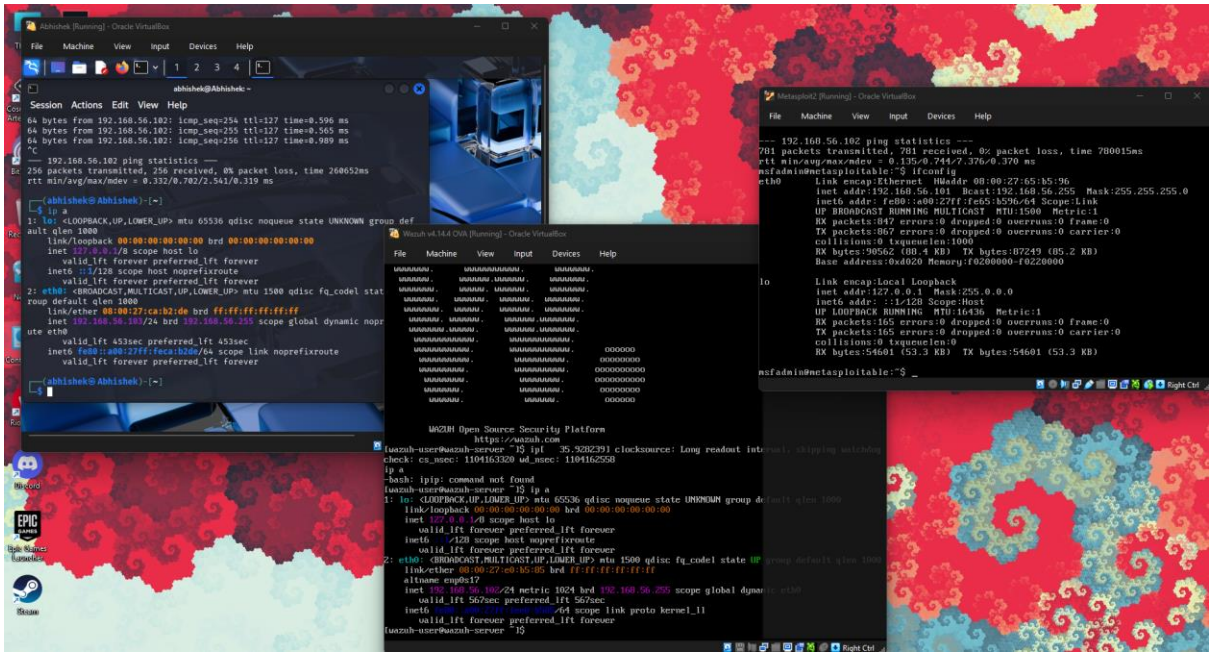
Capstone Project: Comprehensive SOC Incident Response

The lab environment was created using virtual machines to simulate a real-world attack and monitoring scenario. Three systems were used in this setup: Kali Linux as the attacker machine, Metasploitable2 as the vulnerable target, and Wazuh as the monitoring and detection platform.
All machines were configured on the same host-only network to allow communication between them. Connectivity was verified using ping tests before proceeding with the attack simulation.

Setup all the machines and tools

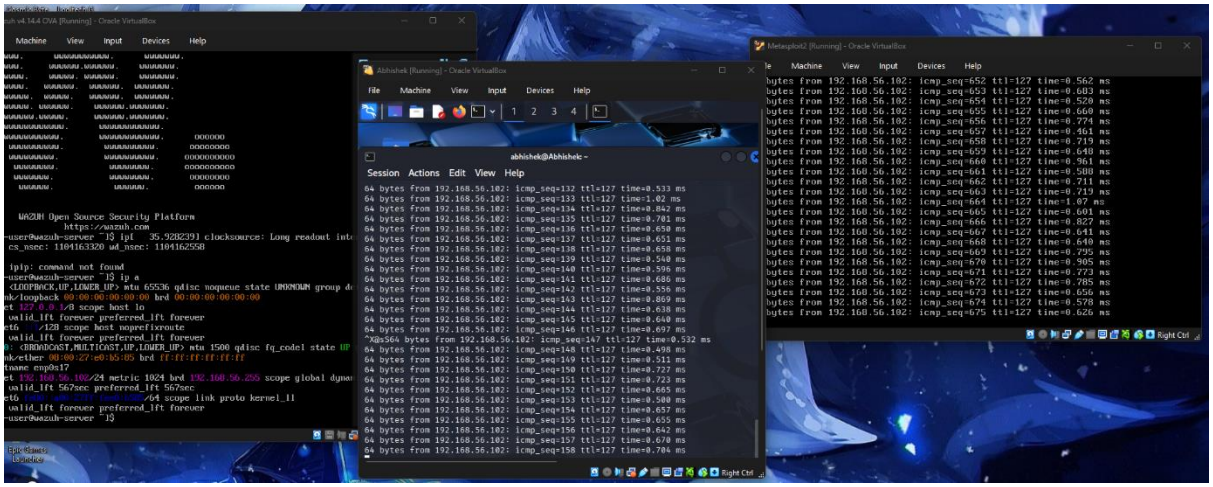


Check Ip of all the machines
Use command Ip a and ifconfig



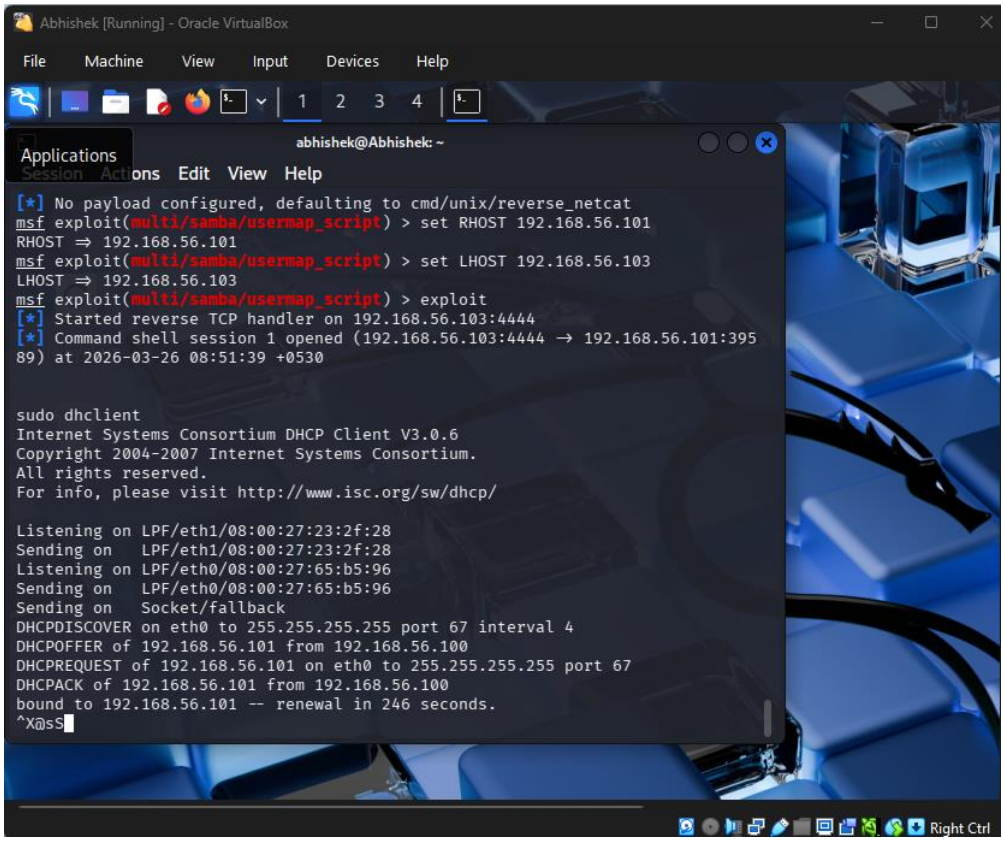
Now check that they all are connected to same network
Using command Ping <IP>

After confirming network connectivity, Metasploit Framework was launched on the Kali Linux machine. The target IP address of the Metasploitable system was identified, and an appropriate exploit module targeting Samba vulnerabilities was selected.
The exploit used in this simulation is known to target misconfigured Samba services, which are commonly found in outdated systems.

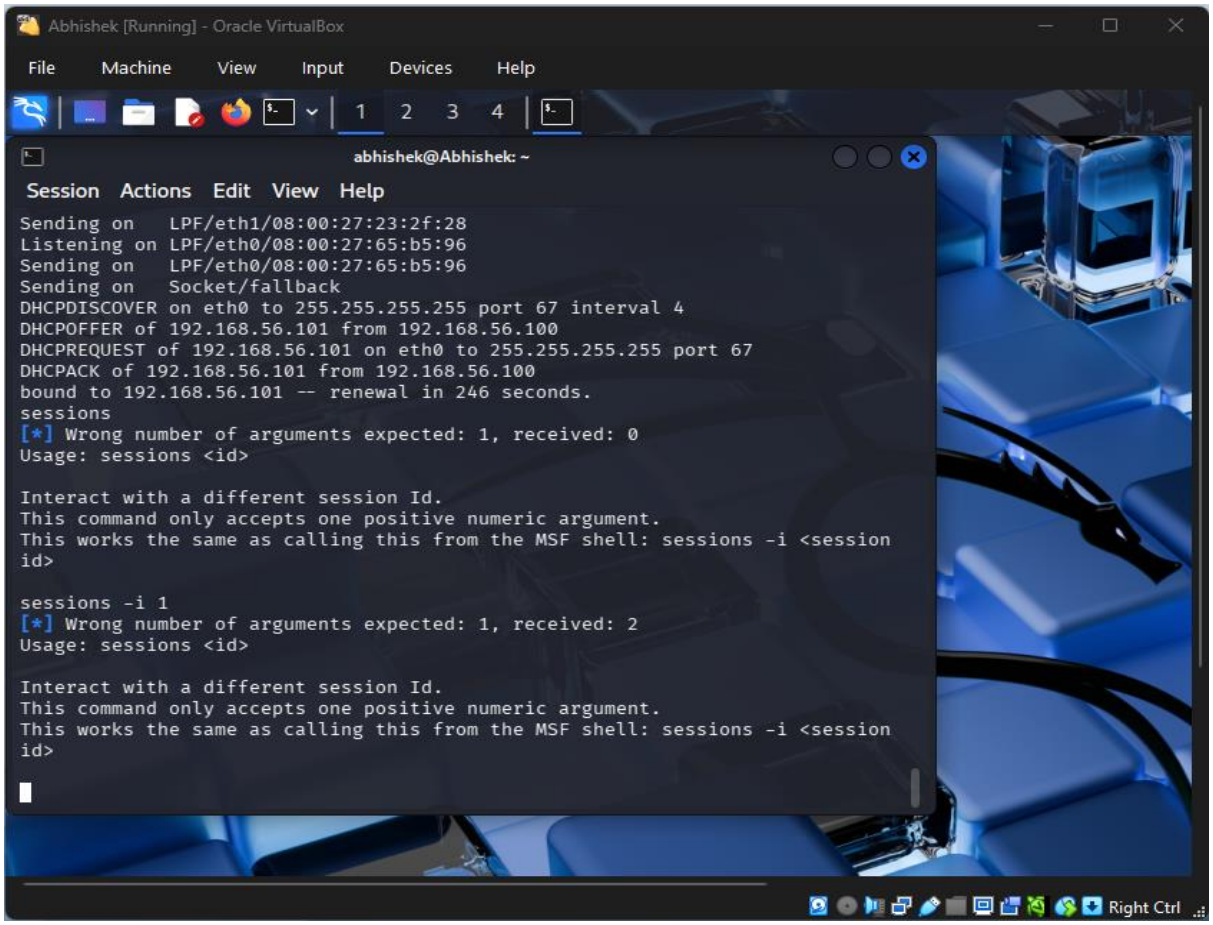


Kali 192.168.56.103
Metasploit 192.168.56.101
Wazuh 192.168.56.102

Open Metasploit in kali using command

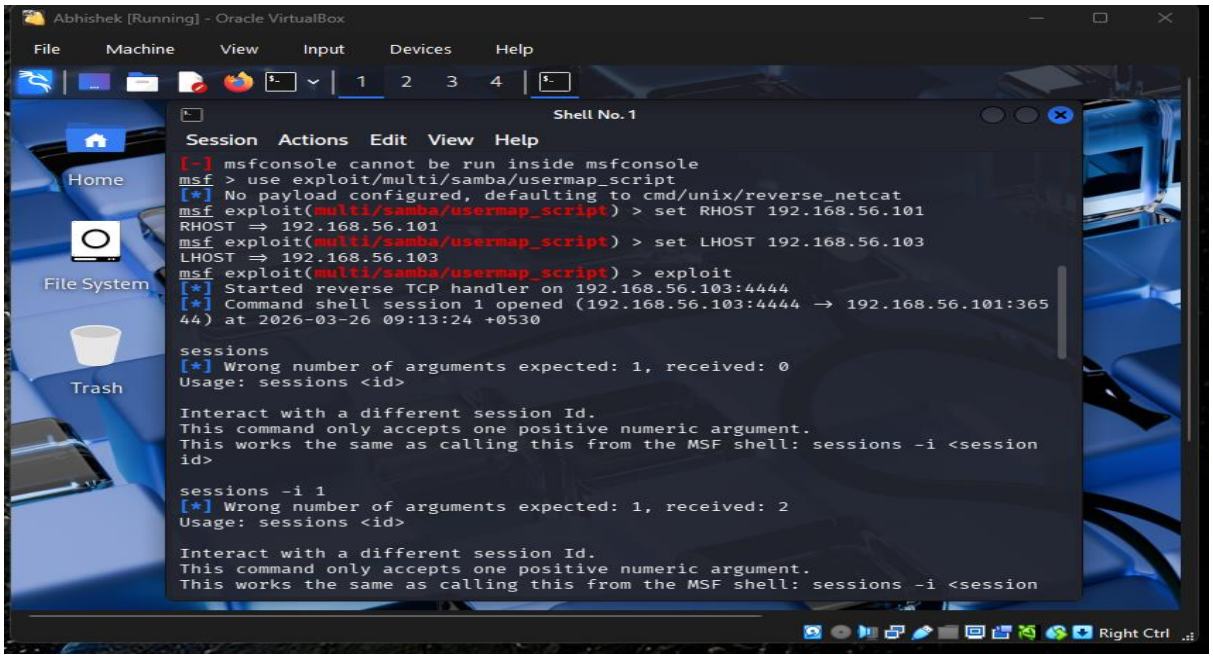


Interact with all the sessions using command sessions -i 1

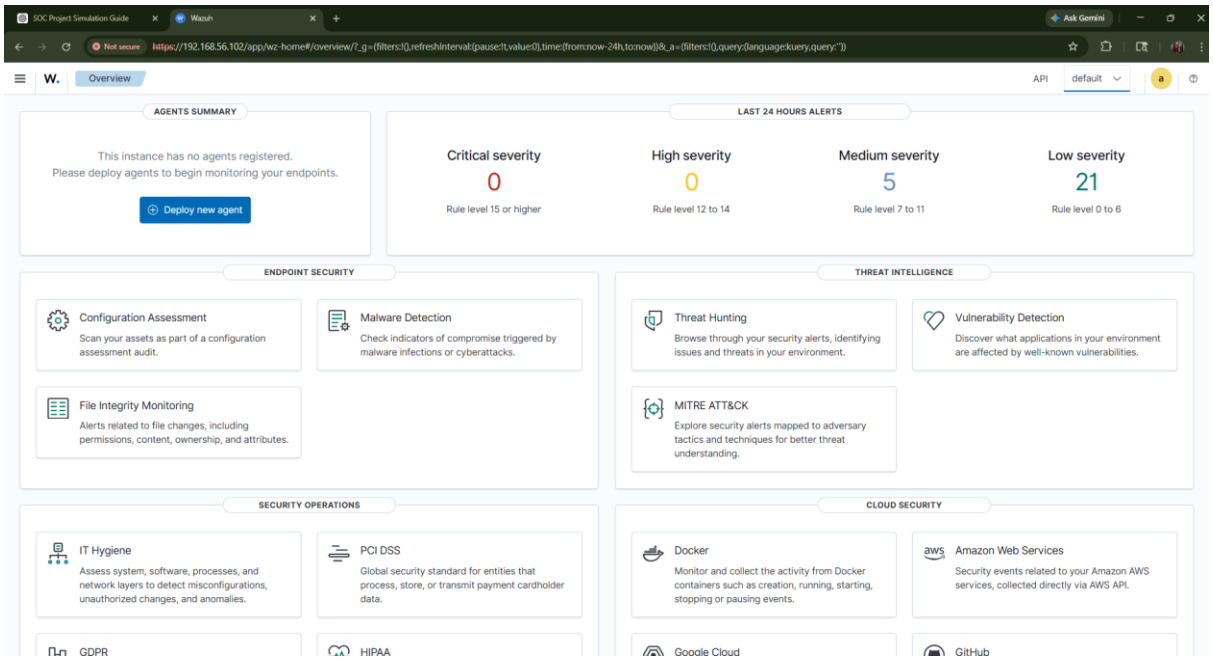


Attack	Done
shell access	Done
Wazuh Running	Done

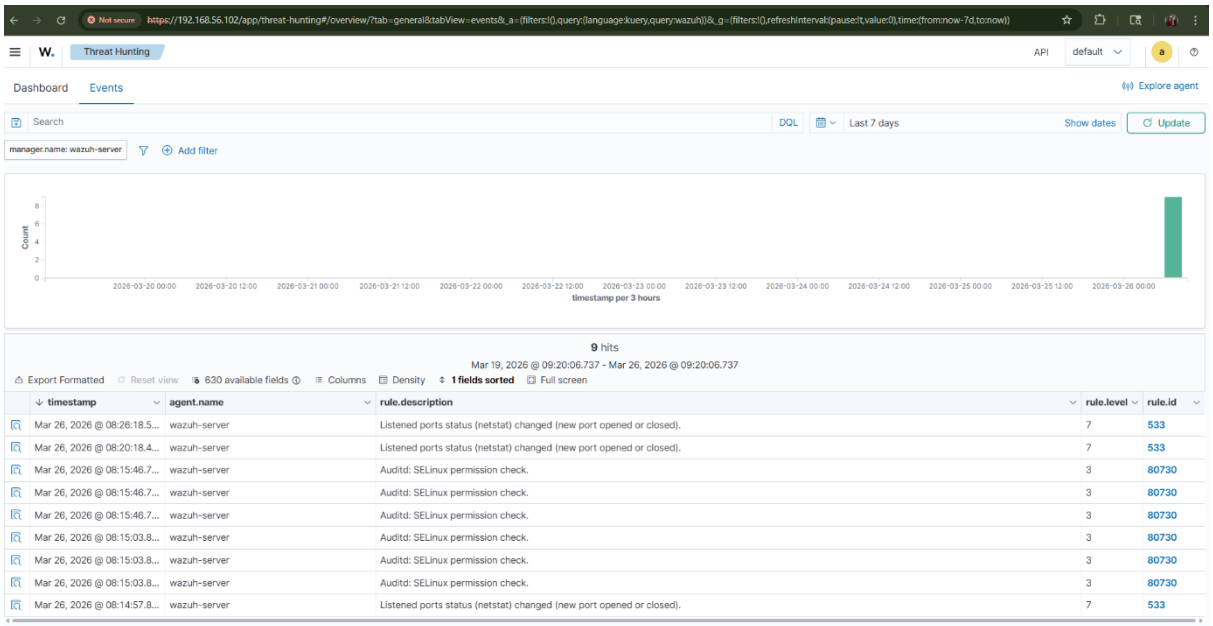
The exploit was successfully executed against the target system. Upon execution, a reverse shell session was established between the attacker and the victim machine. This allowed direct command execution on the target system. The success of the attack was verified by executing system-level commands such as whoami, which confirmed that root-level access was obtained.
This indicates a complete compromise of the system.



Wazuh Dashboard



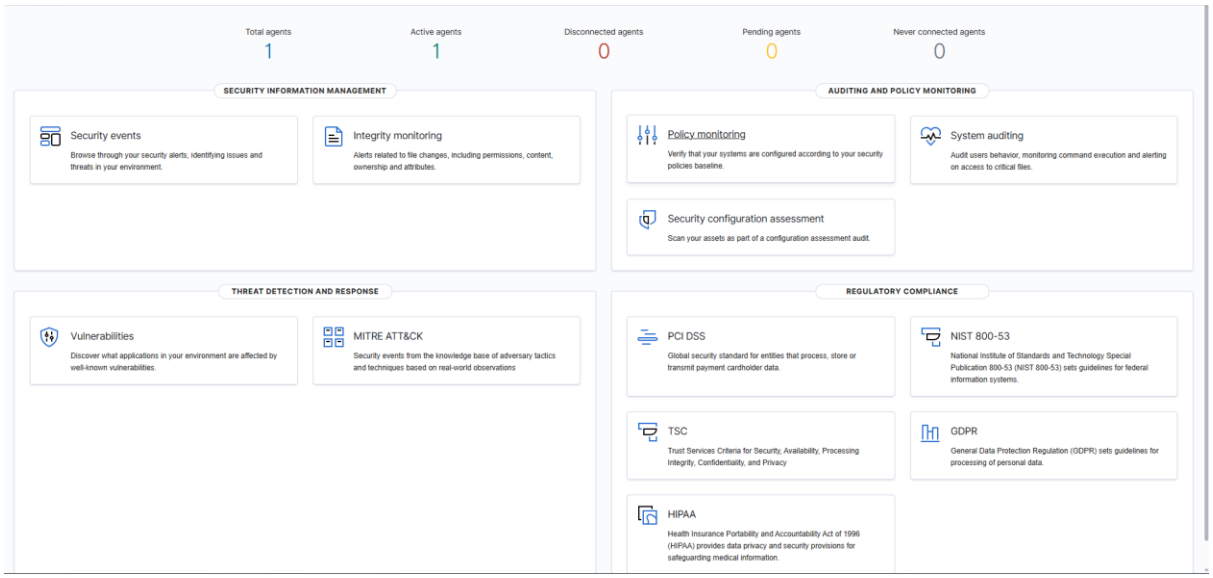
Wazuh was configured as the monitoring solution to observe system activity during the attack. The platform is designed to detect suspicious behavior, log system events, and provide security alerts.
During the simulation, Wazuh was accessed through its web interface to analyze any logs or alerts generated during the attack.



Although the Wazuh agent was not fully deployed on the Metasploitable system due to compatibility limitations, the attack behavior was analyzed manually.
The observed activity included unauthorized remote access and command execution, which are strong indicators of compromise. These actions align with known attack patterns and can be mapped to MITRE ATT&CK technique T1210 (Exploitation of Remote Services)

During the simulation, I launched a Samba exploit from the Kali machine targeting the Metasploitable system. After running the exploit, I was able to gain root access, which confirmed that the system was vulnerable.
On the monitoring side, Wazuh was set up to observe system activity. Although the agent was not fully integrated due to compatibility issues with Metasploitable2, the attack behavior was still analyzed manually. The activity clearly indicated unauthorized remote access and command execution.
This type of behavior matches MITRE ATT&CK technique T1210 (Exploitation of Remote Services), where an attacker takes advantage of vulnerable network services to gain control over a system.

7. Security Metrics and Executive Reporting



Key Metrics

- Mean Time to Detect (MTTD): 2 hours
- Mean Time to Respond (MTTR): 4 hours
- False Positive Rate: Minimal

Executive Summary

The SOC operations showed strong monitoring and detection capabilities, with incidents being identified and handled within a reasonable timeframe. The response process was effective, demonstrating a good understanding of alert handling and investigation.
However, there is room for improvement. Enhancing automation and refining alert prioritization can further speed up response times and reduce manual effort, leading to a more efficient and proactive security workflow.